

CHAPITRE I

DIODES

1. Introduction and overview

This chapter introduces the diode, one of the most important components in electronics, and explains the semiconductor principles on which it is based. First, we review the basic concepts of semiconductors, including intrinsic semiconductors and extrinsic semiconductors such as P-type and N-type materials. Then we study the formation of the PN junction and understand its behavior when it is unbiased as well as when it is forward or reverse biased. After that, we examine the junction diode, focusing on its current–voltage characteristics, and its approximations. The chapter also introduces the diode behavior in AC circuits. Finally, we study the Zener diode, its characteristics and biasing conditions, and how it can be used for voltage regulation in DC circuits as well as in some AC applications. Through these topics, students will understand how diodes are used to control and regulate electrical signals in electronic systems.

2. Semiconductors

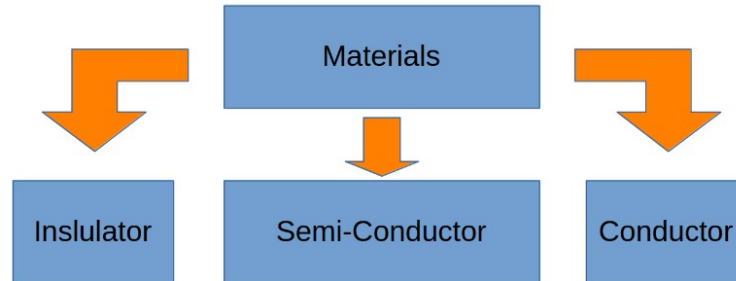
The development of semiconductor technology has enabled the miniaturization of electronic circuits and the creation of highly complex integrated systems, leading to rapid progress in fields such as telecommunications, computing, and cybersecurity.

Semiconductors are the foundation of modern electronics and play a central role in almost every electronic device used today. Unlike conductors and insulators, semiconductor materials have electrical properties that can be precisely controlled through processes such as doping and the application of external electric fields. This controllability allows semiconductors to regulate the flow of electric current, making them ideal for building essential electronic components.

Devices such as diodes, transistors, integrated circuits, and sensors are all based on semiconductor materials, most commonly silicon. These components perform fundamental functions including rectification, amplification, switching, and signal processing. As a result, semiconductors are at the heart of technologies such as computers, smartphones, communication systems, and power electronics.

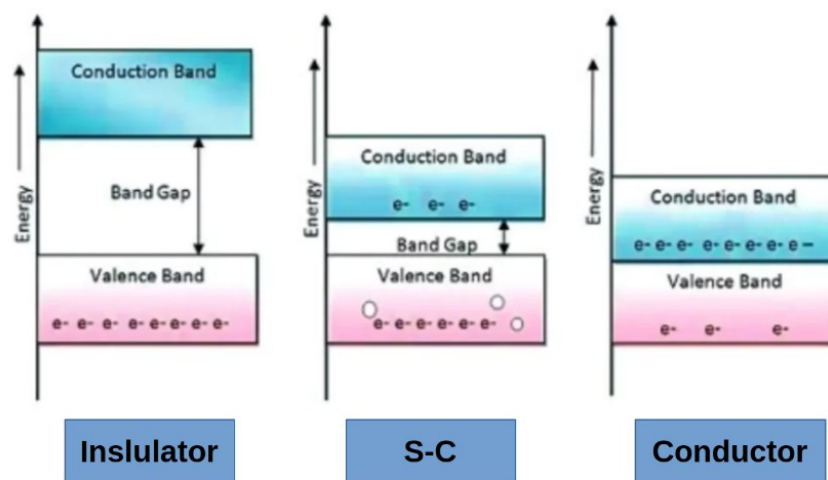
2.1. Classification of materials

Materials can be classified based on their ability to conduct electricity into **conductors**, **insulators**, and **semiconductors**, each having distinct properties.

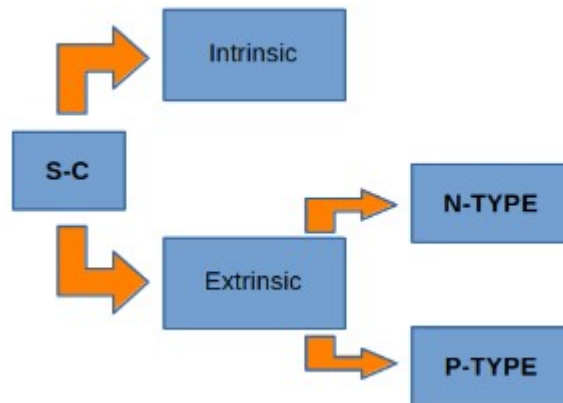


Conductors, such as copper or aluminum, have a very high number of free electrons in their outer energy levels, allowing current to flow easily, they have low resistivity and high conductivity. Insulators, like glass or rubber, have very few free electrons, so they resist the flow of current and have very high resistivity. Semiconductors, such as silicon and germanium, fall between conductors and insulators. Their conductivity is moderate and can be controlled by temperature, impurities (doping), or external voltage.

They can be also classified based on the number of electrons in their valence orbit. Conductors usually have 1 to 3 valence electrons. Insulators have 5 to 8 valence electrons. Semiconductors have 4 valence electrons that form covalent bonds. This difference in valence electrons is one of the key reasons why conductors, insulators, and semiconductors behave so differently.

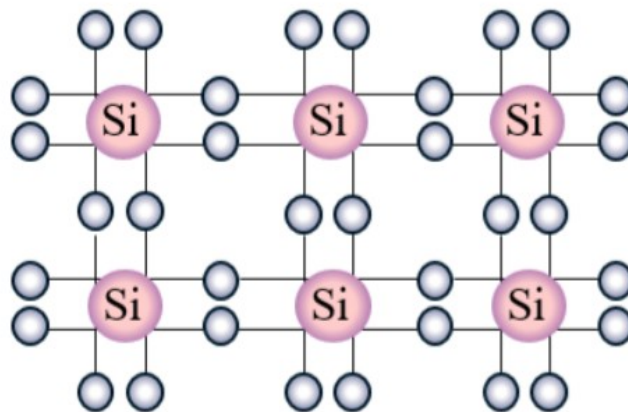


From the energy band perspective, conductors have overlapping valence and conduction bands, insulators have a very wide energy gap more than 3 eV , and semiconductors have a smaller energy gap around 1 eV that allows electrons to move to the conduction band under certain conditions. In terms of applications, conductors are used for wiring, insulators for protection, and semiconductors for electronic devices like diodes and transistors.



2.2. Intrinsic semiconductors

An intrinsic semiconductor is a **pure semiconductor** material without any added impurities. Its electrical conductivity comes only from the electrons naturally present in the material. At absolute zero, an intrinsic semiconductor behaves like an insulator because all electrons are bound in covalent bonds.



When the temperature rises, some electrons gain enough energy to jump from the valence band to the conduction band, leaving behind empty spaces called **holes** that's behaves like a positive charge because the loss of the electron produces a positive ion. The hole will attract and capture any electron in the immediate vicinity. The existence of holes is the critical difference between conductors and semiconductors. Both the free electrons and the

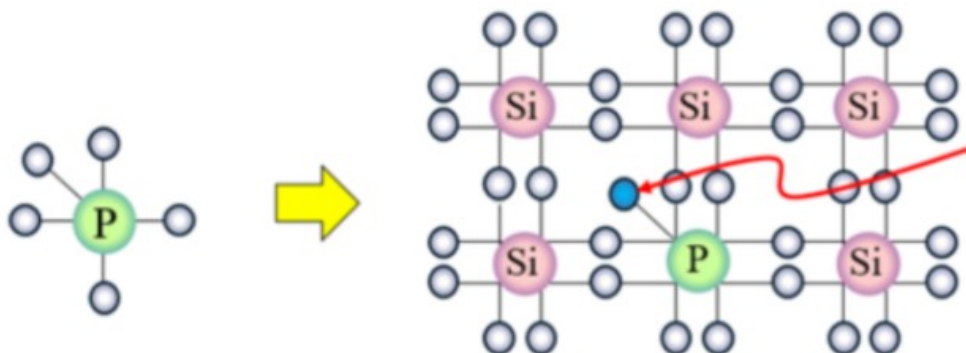
holes can carry electric current. In an intrinsic semiconductor, the number of free electrons is equal to the number of holes, and this balance determines its natural conductivity. Common examples of intrinsic semiconductors are pure silicon and pure germanium.

2.3. Extrinsic semiconductors

The intrinsic semiconductor described above has equal concentrations of free electrons and holes, generated by thermal generation. These concentrations are far too small for silicon to conduct appreciable current at room temperature. Also, the carrier concentrations and hence the conductivity are strong functions of temperature, not a desirable property in an electronic device. Fortunately, a method was developed to change the carrier concentration in a semiconductor crystal substantially and in a precisely controlled manner. This process is known as **doping**, and the resulting silicon is referred to as doped silicon. By adding impurities, we can create two types of extrinsic semiconductors: **N-type** and **P-type**. Extrinsic semiconductors are more conductive than intrinsic ones because the added impurities increase the number of charge carriers, making them essential for devices like diodes and transistors.

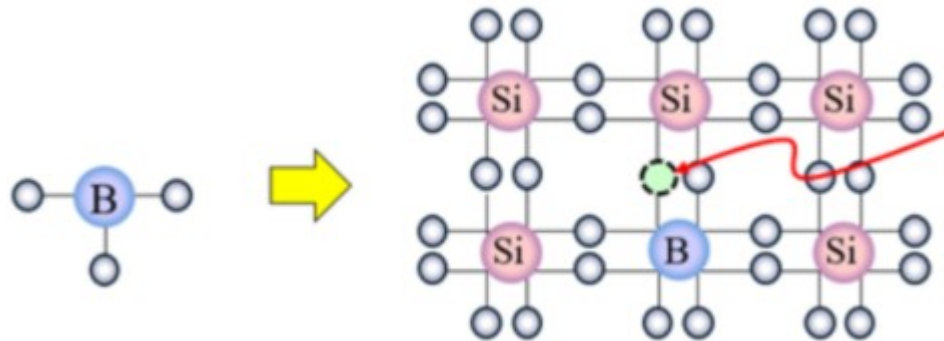
2.3.1. N-type extrinsic semiconductors

An N-type extrinsic semiconductor is created by adding impurities with **five valence electrons** like Phosphorus (called donor atoms) to a pure semiconductor like silicon or germanium. These extra electrons are not bound in covalent bonds and can move freely, increasing the material's conductivity. In N-type semiconductors, electrons are the majority charge carriers, while holes are the minority carriers. The added donor atoms slightly change the electrical properties, allowing the material to conduct current more easily than a pure (intrinsic) semiconductor.



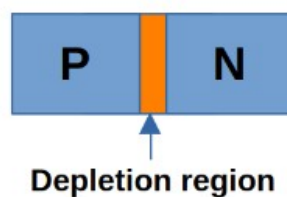
2.3.2. P-type extrinsic semiconductors

A P-type extrinsic semiconductor is made by adding impurities with three valence electrons like the Bore (called acceptor atoms) to a pure semiconductor. These atoms create holes in the crystal structure, which act as positive charge carriers. In P-type semiconductors, holes are the majority carriers, while electrons are the minority carriers. The presence of these holes increases the material's conductivity compared to an intrinsic semiconductor



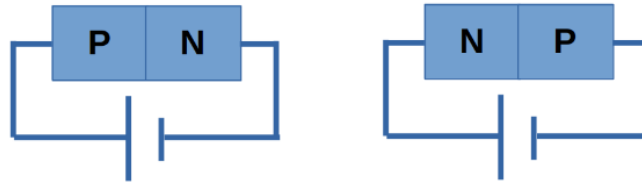
2.4. PN junction

A PN junction is formed when a P-type semiconductor is joined with an N-type semiconductor. At the junction, electrons from the N-side diffuse into the P-side and recombine with holes, creating a region called the **depletion region** where there are no free charge carriers. This region acts as a barrier that prevents further movement of charges under normal conditions.



A PN junction behaves differently depending on whether an external voltage is applied. In an unbiased PN junction, no external voltage is connected, so the depletion region forms naturally at the junction. In a **forward-biased** PN junction, the P-side is connected to the positive terminal and the N-side to the negative terminal of a voltage source, which reduces the barrier and allows current to flow easily. In a **reverse-biased** PN junction, the P-side is connected to the negative terminal and the N-side to the positive terminal, which

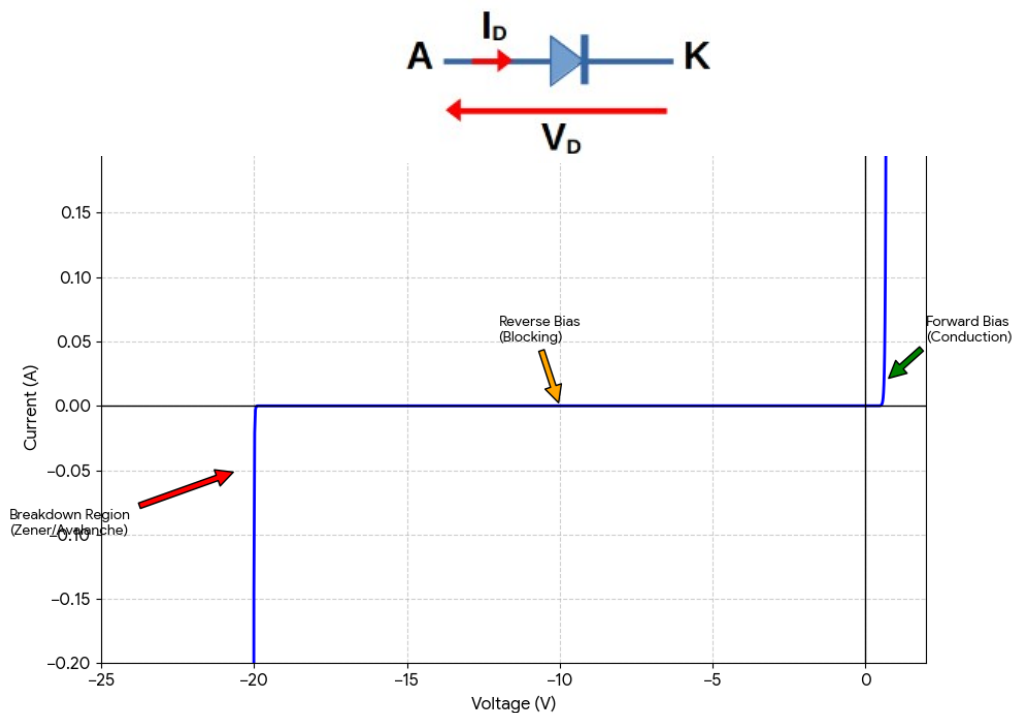
increases the barrier and prevents most current from flowing, except for a very small leakage current.



This difference in behavior is the principle that allows diodes to control the direction of current in circuits.

3. Diodes

A diode is an electronic device that allows current to flow in one direction while blocking it in the opposite direction. Its operation is based on the PN junction. It is two terminal non linear device whose I-V characteristic besides exhibiting non-linear behavior is also polarity dependent. The non-linear, and polarity characteristics of the diode make for a very interesting and useful device.



The mathematical function that describes the relationship between the voltage V_D , and the diode current I_D of a real model diode is as follow :

$$I_D = I_S \left[\exp\left(\frac{V_D}{V_T}\right) - 1 \right]$$

The parameter I_S is called the reverse saturation current and it is independent of the diode voltage V_D (For silicon, $I_S = 10^{-12}$ A). The parameter $V_T = \frac{KT}{q}$ is called the thermal voltage. At room temperature $V_T = 26$ mV (k = Boltzmann's constant, T = the temperature and q = the electronic charge).

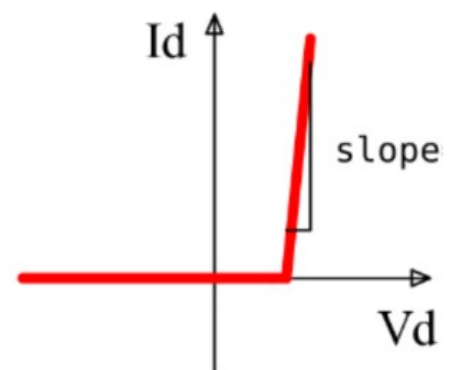
The figure above shows the variation of I_D as function of V_D , it highlights three primary regions of operation. **The Forward Bias** ($V_D > 0$ V), When a positive voltage is applied to the anode relative to the cathode, the diode is "forward-biased." Initially, very little current flows. however, as the voltage reaches the knee/threshold voltage (approximately 0.7V for silicon or 0.3V for germanium), the current increases exponentially. In this region, the diode state is ON. **The Reverse Bias** ($V_D < 0$ V), When the voltage is reversed, the diode is "reverse-biased." The current value is almost negligible. In this region, the diode effectively acts as an open switch (OFF). The last region is **the Breakdown Region** (negligible in the standard model) If the reverse voltage is increased beyond a specific threshold (the Breakdown Voltage), the diode will begin to conduct heavily in the reverse direction. This is the primary operating principle for Zener diodes.

3.1. Diode approximations

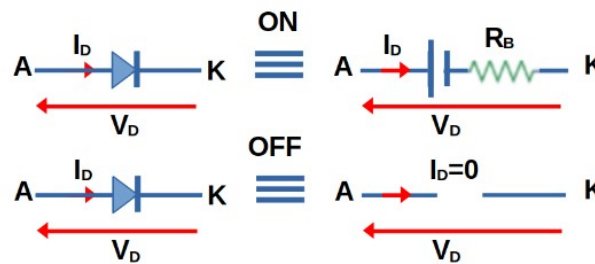
In circuit analysis, we use approximations because the $I_D = f(V_D)$ equation is nonlinear. Since the current is an exponential function of the voltage, calculating precise values in a complex circuit would require iterative numerical methods or specialized software. To simplify this, engineers use three standard models, also called approximations.

3.1.1. Practical diode

This model shows the most accurate of the linear approximations, this model accounts for the fact that semiconductor material has a small amount of internal resistance, known as **bulk resistance** (R_B). Once the $V_D = 0.7$ V

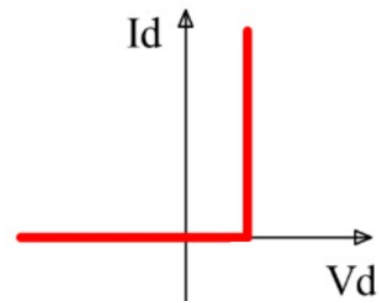
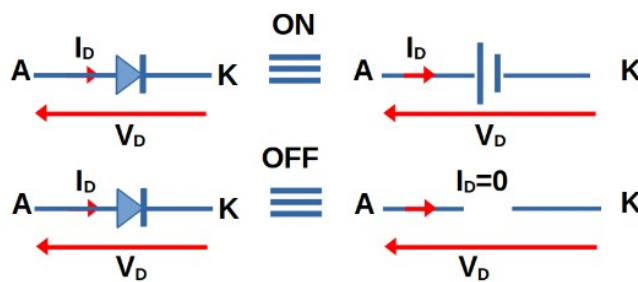


, the current does not rise vertically. Instead, it follows a slope defined by $\frac{1}{R_B}$. In a circuit, the diode will be replaced as follow :



3.1.2. Constant voltage

This model is the most frequently utilized approximation in circuit analysis because it strikes an effective balance between simplicity and realism. Since R_B is very small it may be neglected ($R_B = 0$, slope = ∞).

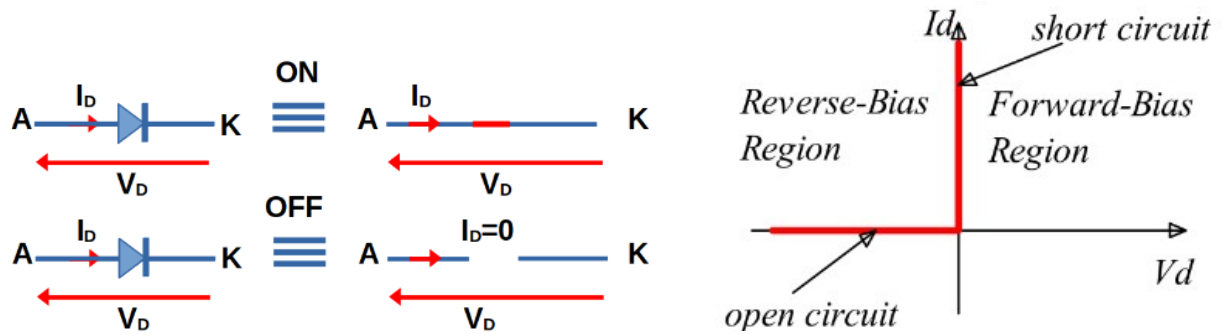


In the model the diode is treated as a component that requires a specific threshold of energy to begin conduction. When forward-biased, the diode is modeled as a fixed voltage source (typically **0.7V** for Silicon or **0.3V** for Germanium) connected in series with an ideal switch. This means that if the applied source voltage is less than this barrier potential, the diode remains non-conductive (**D:OFF**), once the voltage meets or exceeds this value, the diode "turns on" and maintains a constant voltage drop regardless of the amount of current flowing through it. Conversely, in reverse bias, the model assumes the diode acts as a perfect insulator ($I_D = 0$).

3.1.3. Ideal diode

The Ideal Model is the most basic approximation of a diode, it is assumed to have zero resistance when forward-biased, acting as a **short circuit** (D : ON) that allows current to flow freely with no voltage drop across its terminals. When reverse-biased, it is treated as having

infinite resistance, acting as a perfect **open circuit** that completely blocks any flow of current ($I_D=0$), regardless of the magnitude of the reverse voltage.



This "on-off" binary behavior ignores the 0.7 V or 0.3 V (according to the SC type) barrier potential and internal resistance found in physical semiconductors, making it an excellent tool for initial circuit troubleshooting or for analyzing high-voltage systems where small voltage drops are mathematically insignificant.

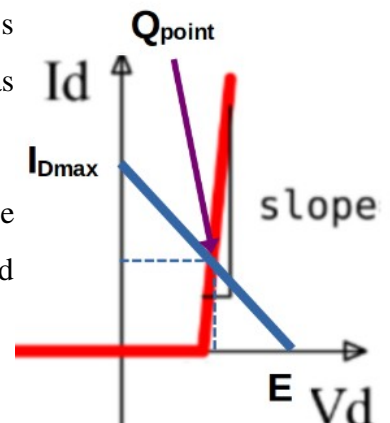
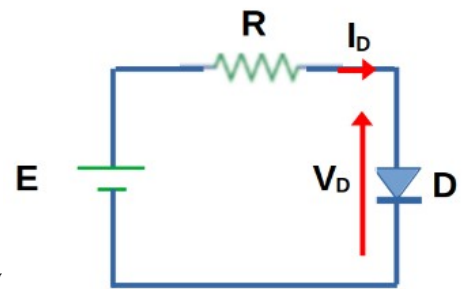
3.2. The Load Line and Q-point

The Load Line ($I_D=f(V_D)$) is a graphical tool used in circuit analysis to determine the exact operating state of a diode when it is connected to a specific **voltage source** and **resistor**. While a diode's own characteristic curve describes its internal physical behavior according to its approximation, the load line represents the constraints imposed by the external circuit according to Kirchhoff's Voltage Law.

$$E = R I_D + V_D \Rightarrow I_D = -\frac{1}{R} V_D + \frac{E}{R}$$

By plotting the circuit's linear equation (defined by the source voltage and the resistance) directly over the diode's exponential curve, we can identify a unique intersection known as the Quiescent Point or Q-point.

This point is significant because it reveals the precise simultaneous value of the current flowing through the circuit and the voltage drop across the diode for that specific configuration.

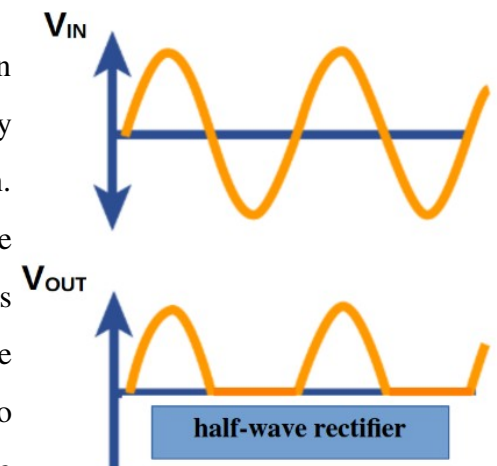


3.3. Diode operations in AC

In AC circuits, diodes serve as the fundamental building blocks for signal processing and power conversion. Beyond simple rectification, they are used also to modify the shape of a waveform or capture specific voltage levels. The AC Diode circuit analysis typically begins by assuming the device is in a conducting (**ON**) state. We then solve for the input voltage conditions required to maintain this state. If these conditions are not met, the diode is considered to be in the non-conducting (**OFF**) state. The final step is to determine the resulting output voltage for each situation.

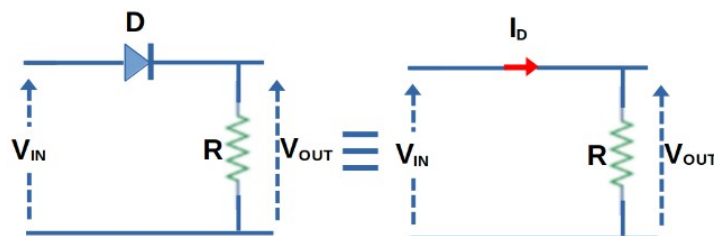
3.3.1. The Rectifier

A rectifier is a fundamental circuit used to convert an alternating current (AC) into a direct current (DC) by leveraging the diode's ability to conduct in only one direction. In a half-wave rectifier, a single diode blocks the negative half-cycles of the AC input, allowing only the positive peaks to reach the load. To improve efficiency, a full-wave bridge rectifier uses four diodes arranged in a diamond pattern to redirect the negative half-cycles so they also flow in the positive direction.



Example : Circuit analysis steps

- Fix the diode approximation (considered Ideal here),
- Assume that the D is turned ON, and replace it with the equivalent circuit,

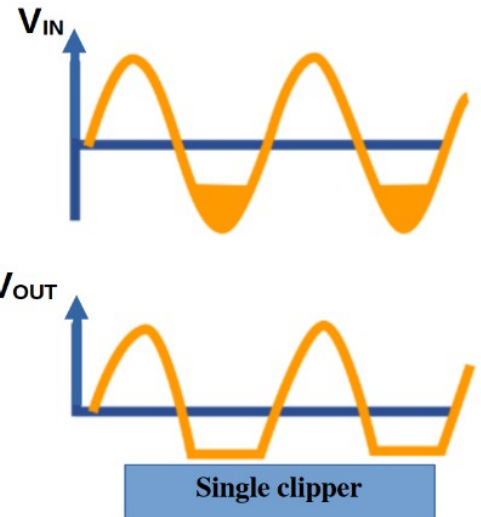


- Determine the I_D expression, $I_D = \frac{V_{Input}}{R} > 0$
- Determine both conditions for conducting and non-conduction of the diode as function of V_{Input} ,
- Calculate V_{Out} for the different situations.

3.3.2. The Diode Clipping Circuit (Limiters)

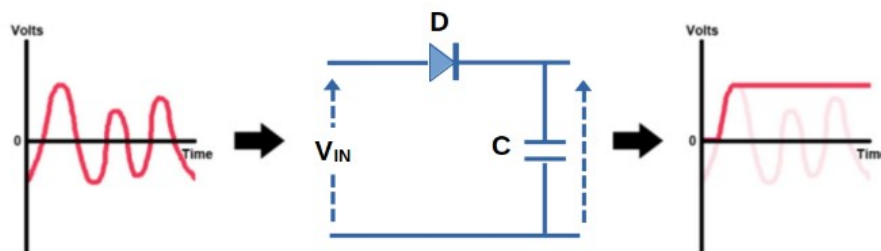
A Clipping Circuit is used to cut off or "limit" a portion of an AC waveform that exceeds a certain voltage threshold, protecting sensitive electronic components downstream. In this configuration, a diode is placed in parallel with the load resistor. When the AC signal voltage is below the diode's threshold, the diode acts as an open switch, and the signal passes to the output unchanged. However, as soon as the signal voltage exceeds the diode's turn-on voltage (or a connected DC bias battery), the diode conducts and acts as a short circuit.

This caps the output voltage at that specific threshold, flattening or "squaring off" the peaks of the sine wave.



3.3.2. The Peak Detector

A peak detector is a specialized circuit designed to capture and hold the maximum instantaneous voltage level of an analog signal. It consists of a diode connected in series with a capacitor; as the input voltage rises, the diode becomes forward-biased and charges the capacitor to the peak value. When the input signal begins to drop, the diode becomes reverse-biased, effectively "locking" the charge inside the capacitor.



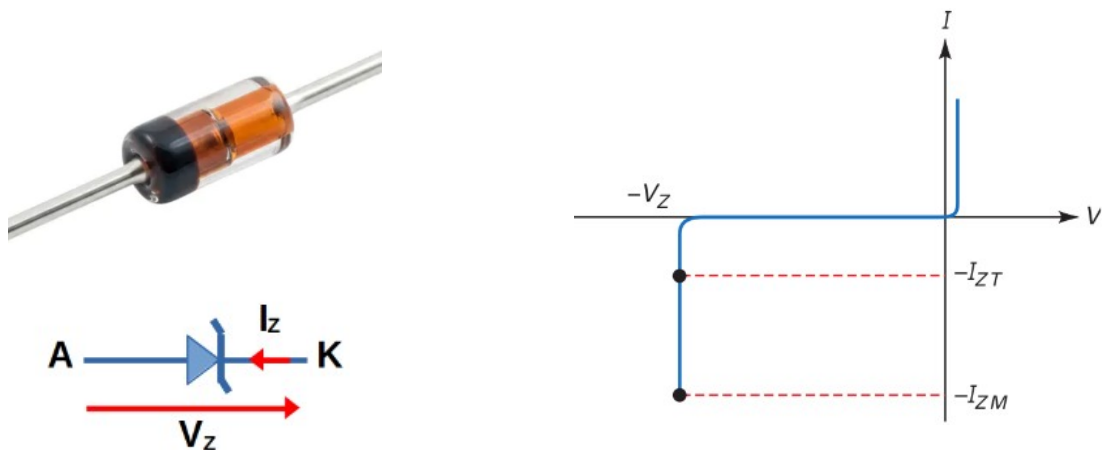
This allows the circuit to produce a DC output that tracks the highest point of the input waveform, a function that is critical for measuring signal strength or extracting audio from radio waves.

Beyond rectifiers, clippers, and peak detectors, diodes are essential in several other specialized signal-processing circuits. In high-voltage applications, **voltage multipliers** combine diodes and capacitors in a ladder network to step up an AC input to a much higher DC output voltage. Additionally, diodes are used in **switching circuits** and **logic gates** (such as AND and OR gates) to perform digital operations, as well as in **protection circuits** to divert transient voltage spikes away from sensitive components. These diverse applications

demonstrate that the diode's primary function—controlling the direction of current—is the building block for both power conversion and complex signal manipulation.

4. Zener Diode

While a standard rectifier diode is designed to block current in the reverse direction, the Zener diode is specially doped to operate safely in the **breakdown region**. As we know, when normal diodes operate in the breakdown region, they usually experience thermal runaway, where the heat produced by high current damages or destroys the semiconductor material. In contrast, Zener diodes are designed to handle this condition safely. Once the reverse voltage reaches a specific value, called the **Zener voltage** (V_Z), the diode enters breakdown in a controlled manner. It then allows current to flow while maintaining a very stable voltage across its terminals.



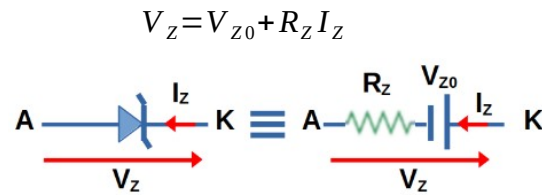
In the forward region, the Zener diode begins to conduct at about 0.7 V, just like a normal silicon diode. In the leakage region (between zero voltage and the breakdown voltage), only a very small reverse current flows. When the Zener diode reaches the breakdown region, the curve shows a very sharp knee, followed by a rapid increase in current. However, the voltage remains almost constant, and it is approximately equal to the Zener voltage over most of the breakdown region. Manufacturers usually specify the value of V_Z at a particular test current called I_{ZT} in the diode data sheet.

4.1. Zener diode approximations

The Zener approximation is a simple method used to analyze circuits that contain a Zener diode. Mainly, we have two categories : **Practical** and **Ideal**

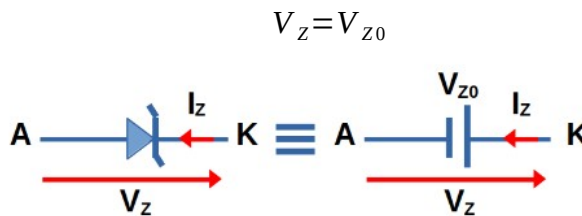
4.1.1. Practical Zener diode

In this type, the Zener diode can be modeled as an ideal voltage source in series with a small resistance called the Zener resistance (R_Z). The total voltage across the Zener diode is equal to the breakdown voltage plus the small voltage drop across this resistance. Because the value of R_Z is usually very small, its effect on the total voltage across the Zener diode is minor, and the voltage remains nearly constant.



4.1.2. Ideal Zener diode

the Zener diode is treated as an ideal voltage source ($R_Z=0$) when it operates in the breakdown region. This means that the voltage across the diode is assumed to remain constant and equal to the Zener voltage (V_Z , regardless of changes in current.

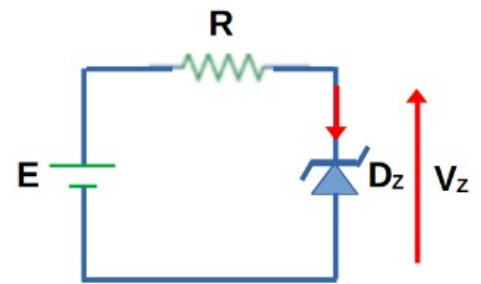


4.2. The Load Line and Q-point in Zener

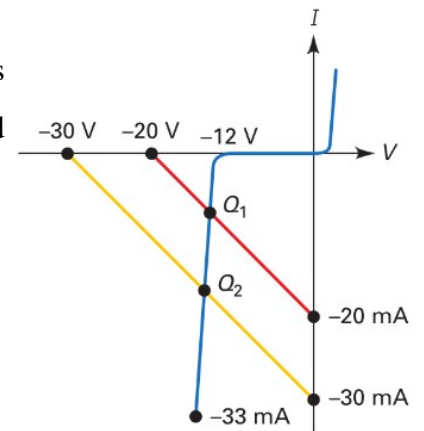
The load line and Q-point are important tools used to analyze the operation of a Zener diode in a voltage regulator circuit. The load line ($I_Z = f(V_Z)$) represents all the possible combinations of current and voltage in the circuit, based on the supply voltage and the series resistance. It is drawn on the same graph as the Zener diode characteristic curve. The point where the load line intersects the Zener characteristic curve is also called the Q-point. This point shows the actual operating voltage and current of the Zener diode under steady conditions. A properly chosen Q-point ensures that the diode remains in the breakdown region and provides stable voltage regulation.

$$E = V_R + V_Z \Rightarrow E = R I_Z + V_Z$$

$$I_Z = -\frac{1}{R} V_Z + \frac{E}{R}$$



Suppose $E = 20\text{ V}$ and $R = 1\text{ k}\Omega$. Then, the corresponding load line is in **RED**. As result, we get Q_1 . If $E = 30\text{ V}$, we obtain a novel load line in **YELLOW** and even a novel Q_2



4.3. Zener diode operations in AC

When a Zener diode is used with an AC voltage source, it usually works together with a series resistor to limit the current. During the positive half-cycle of the AC signal, the diode behaves like a normal forward-biased silicon diode, conducting when the voltage exceeds about 0.7 V. During the negative half-cycle, the diode becomes reverse-biased. If the reverse voltage reaches the Zener voltage (V_Z), the diode enters breakdown and conducts in the reverse direction, maintaining a nearly constant voltage across its terminals. This property allows Zener diodes to be used for AC voltage clipping, limiting, or for simple voltage regulation after rectification. **For circuit's analysis, the same method used for the normal diode is adopted.**